Electronics and Telecommunications Engineering: (Machine learning project)

Problem statement No.40 – Network Intrusion Detection System

The Challenge: Create a robust network intrusion detection system (NIDS) using machine learning. The system should be capable of analysing network traffic data to identify and classify various types of cyber-attacks (e.g., DoS, Probe, R2L, U2R) and distinguish them from normal network activity. The goal is to build a model that can effectively secure communication networks by providing an early warning of malicious activities.

Kaggle dataset link – https://www.kaggle.com/datasets/sampadab17/network intrusion-detection
Technology – Use of IBM cloud lite services is mandatory.